Pump P-101 Maintenance Report

1. Document Information

Equipment Tag: P-101
Equipment Type: Centrifugal Process Pump
Service: Industrial process-fluid transfer
Report Type: Maintenance History and Inspection Report
Document Status: Synthetic Demonstration Document
Revision: 1.0

Note: This document contains synthetic maintenance information created for the Sovereign AI Workbench demonstration. It does not represent a real industrial maintenance record.

---

2. Equipment Details

Equipment: Pump P-101
Rated Speed: 1450 RPM
Rated Flow: 120 m³/h
Motor Power: 30 kW
Normal Vibration: Below 4.5 mm/s RMS
Warning Vibration: 4.5–7.1 mm/s RMS
Critical Vibration: Above 7.1 mm/s RMS

---

3. Maintenance History

Maintenance Event 1

Date: 12 March 2026
Maintenance Type: Routine Preventive Maintenance

Observations

- Pump casing was inspected and found to be in acceptable condition.
- Coupling was visually inspected.
- Bearing lubrication was checked.
- Mechanical seal showed no significant leakage.
- Vibration reading was **2.6 mm/s RMS**.
- Bearing temperature was **64 °C**.
- Discharge pressure was **4.2 bar**.
- Pump speed was **1450 RPM**.

Actions Performed

- Bearing lubrication was renewed.
- Coupling fasteners were checked.

- Mechanical seal was inspected.
- Pump mounting bolts were checked.
- General housekeeping around the equipment was completed.

Result

The equipment was returned to normal operation.

---

4. Maintenance Event 2

Date: 18 May 2026
Maintenance Type: Condition-Based Inspection

Trigger

The condition-monitoring system showed a gradual increase in pump vibration.

Recorded vibration: 5.3 mm/s RMS

The value exceeded the normal operating range but remained below the critical limit.

Inspection Findings

- Slight coupling misalignment was observed.
- Bearing condition was acceptable but showed early signs of wear.
- No significant mechanical seal leakage was observed.
- Pump mounting bolts were secure.

Corrective Actions

- Coupling alignment was corrected.
- Bearing lubrication was checked and replenished.
- Vibration was measured after corrective action.

Post-Maintenance Readings

Vibration: 3.1 mm/s RMS
Temperature: 66 °C
Pressure: 4.2 bar
Speed: 1450 RPM

Result

Vibration returned to the normal operating range.

---

5. Maintenance Event 3 — High Vibration Incident

Date: 21 July 2026
Maintenance Type: Corrective Maintenance

5.1 Problem Reported

During normal operation, the monitoring system detected a significant increase in vibration.

Sensor Readings

Parameter	**Reading**	**Condition**
Vibration	**8.7 mm/s RMS**	Critical
Temperature	71 °C	Elevated
Discharge Pressure	4.1 bar	Normal
Speed	1450 RPM	Normal

The vibration exceeded the critical limit specified in the equipment manual.

5.2 Initial Response

The abnormal reading was verified using the monitoring system.

The operator was notified and the equipment was placed under controlled investigation according to the applicable operating procedure.

Recent maintenance history was reviewed because P-101 had previously experienced increasing vibration associated with coupling alignment.

5.3 Inspection Findings

A detailed mechanical inspection identified the following conditions:

- Increased bearing wear was observed.
- Coupling alignment was outside the acceptable alignment condition.
- No major casing damage was identified.
- Mechanical seal condition was acceptable.
- Pump mounting bolts were secure.
- Impeller condition was visually acceptable.

The investigation concluded that **bearing degradation and coupling misalignment were the primary suspected contributors to the high vibration condition**.

5.4 Corrective Actions

The following maintenance actions were performed:

1. Pump was safely isolated according to the applicable procedure.

2. The affected bearing was removed and replaced.
3. Coupling alignment was corrected.
4. Coupling fasteners were inspected and secured.
5. Bearing lubrication was completed.
6. Pump rotation was checked.
7. The pump was restarted under controlled conditions.
8. Vibration and temperature were monitored after restart.

5.5 Post-Maintenance Readings

Parameter	Before Maintenance	After Maintenance
Vibration	8.7 mm/s RMS	2.6 mm/s RMS
Temperature	71 °C	65 °C
Discharge Pressure	4.1 bar	4.2 bar
Speed	1450 RPM	1450 RPM

The vibration returned to the normal operating range after corrective maintenance.

---

6. Root Cause Assessment

Based on the inspection findings, the most likely contributors to the high vibration event were:

- Bearing degradation
- Coupling misalignment

The sensor data indicated the abnormal condition, while the physical inspection provided the information required to identify the likely mechanical causes.

The sensor reading alone should not be treated as sufficient evidence to establish a root cause.

---

7. Preventive Actions

To reduce the possibility of recurrence:

- Monitor P-101 vibration regularly.
- Review vibration trends rather than relying only on individual readings.
- Inspect coupling alignment during scheduled maintenance.
- Monitor bearing condition and lubrication.
- Investigate sustained increases in vibration.

- Maintain accurate maintenance records.
- Verify abnormal sensor readings before taking major corrective action.

---

8. Maintenance Recommendations

The following checks are recommended during future inspections:

Mechanical Checks

- Bearing condition
- Coupling alignment
- Shaft condition
- Impeller condition
- Pump mounting
- Mechanical seal

Condition Monitoring

- Vibration
- Bearing temperature
- Discharge pressure
- Flow
- Motor speed

Any sustained deviation from the normal operating range should be investigated according to the applicable SOP.

---

9. Relationship to Sensor Analytics

The Sovereign AI Workbench uses sensor data to identify abnormal equipment behavior.

For the July 21, 2026 event:

Vibration: 8.7 mm/s RMS

The equipment manual defines values above **7.1 mm/s RMS** as critical.

The Data Analytics module can therefore identify the event as an abnormal high-vibration condition.

The maintenance report provides historical context and records the subsequent inspection and corrective actions.

---

10. Final Equipment Condition

After bearing replacement and coupling realignment:

Vibration: 2.6 mm/s RMS
Temperature: 65 °C
Discharge Pressure: 4.2 bar
Speed: 1450 RPM

P-101 was returned to normal operating condition following corrective maintenance.

Continued condition monitoring is recommended.

---

11. Document Control

Equipment: P-101
Document: Maintenance History and Inspection Report
Revision: 1.0
Purpose: Synthetic demonstration and AI/RAG testing
Record Type: Demonstration data only

End of Document